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Quantifying the Relationship between Socioeconomic Status and Parent-Child Attachment on Adolescents through Machine Learning

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Abstract—**Parent-child relationships have a significant impact on adolescent development, influencing social cohesion, intellectual growth, and emotional well-being. Adolescence entails rapid biological, cognitive, and neurodevelopmental changes that profoundly impact psychosocial functioning and interpersonal dynamics. Socioeconomic status (SES) is a crucial factor influencing parent-child relationships, although quantifying how parent-child attachment influences SES remains challenging. To bridge this gap, we propose a Machine Learning (ML) framework to classify SES based on parent-child attachment factors, by using the publicly available dataset, which consists of survey data from 239 families (478 samples). Following rigorous preprocessing including Principal Component Analysis (PCA) for feature selection, z-score normalization, outlier mitigation, and missing value imputation, seven classifiers (Decision Tree, Random Forest, k-Nearest Neighbors, Support Vector Machine, AdaBoost, XGBoost, and Logistic Regression) were trained and evaluated using k-fold cross-validation. Upsampling techniques were used to reduce class imbalance, which significantly enhanced model performance. XGBoost demonstrated the effectiveness of ensemble approaches in SES classification by attaining a top accuracy of 89.00% and an AUC-ROC of 98.40%, followed by Random Forest with 88.40% and Decision Tree with 87.33%. Our findings indicate that attachment dimensions, particularly trust, communication, and alienation in both maternal and paternal relationships, are closely associated with SES categories. These findings provide a data-driven approach for creating focused interventions to improve the well-being of adolescents and reduce socioeconomic inequities.**

Index Terms—**Socioeconomic Status (SES), Parent-Child Relationships, Machine Learning, Principal Component Analysis (PCA), XGBoost, Adolescent development.**

I. INTRODUCTION

Adolescence represents a critical developmental period characterized by profound physical, cognitive, emotional, and social transformations [1]. SES plays a central role during this period, influencing education, mental health, and overall well-being [2], [3]. Commonly measured through parental income, education, and occupation, SES shapes the resources and environment available to adolescents both at home and in school [4], [5]. A substantial body of research shows that adolescents from lower SES backgrounds often face limited access to educational resources, reduced parental engagement, and higher levels of financial strain. These constraints can hinder emotional resilience and restrict opportunities for intellectual growth [6], [7]. Understanding how SES shapes adolescents' social and psychological adaptation is therefore critical for designing equitable interventions that promote educational success and mental well-being. One of the strongest links between SES and developmental outcomes lies in the quality of the parent–child relationship [8]. When parents provide warmth, stability, and emotional security, adolescents are more likely to excel academically, cultivate strong social skills, and develop a healthy sense of self-worth [9], [10]. On the other hand, financial difficulties can cause parents to be inconsistent or harsh, which can lead to more emotional struggles and behaviors like defiance or aggression [11], [12]. Strained family relationships may also diminish adolescents' interest in school and heighten susceptibility to negative peer influences, compounding the disadvantages associated with low SES [13], [14]. Economic hardship can also limit parents' capacity to participate in their child's education, whether through reduced time, diminished energy, or fewer available resources [5], [15]. However, research consistently highlights that authoritative parenting, marked by warmth, structure, and clear expectations, can buffer many of the risks linked to socioeconomic disadvantage and support healthy emotional and cognitive development across SES levels [4], [16]. Importantly, financial stability benefits not only children but also parents' mental health, enabling consistent and supportive engagement [6], [17]. When such stability is present, adolescents tend to show stronger motivation, better self-regulation, and greater academic persistence [3]. Conversely, those from low SES households may take on more domestic responsibilities and navigate stressful home environments, both of which can detract from study time and academic engagement [5], [7]. Getting involved in extracurricular activities, especially those related to arts and creative outlets, has been found to be a valuable protective factor, helping improve academic outcomes and providing emotional relief [2], [18]. Similarly, strong peer support and social networks play an important role in improving school engagement, self-esteem, and behavioral adjustment [19], [20]. Community programs and mentorship opportunities also play a vital role in reducing the psychological effects of economic hardship [20], [21].

Adolescents' socioeconomic outcomes are influenced by multiple familial and environmental factors. While prior social-science-based models primarily relied on demographic and survey-based analyses, our study introduces a novel ML framework to classify SES based on parent-child attachment dimensions. By leveraging seven classifiers and addressing class imbalance, our framework uncovers complex, non-linear relationships between attachment factors and SES, providing deeper empirical insights that were previously unexplored.

Considering the interplay between SES, parent–child relationships, and adolescent development, this study examines the socioeconomic pathways affecting academic achievement and psychological well-being. We analyze the mediating effects of parenting style, attachment, and extracurricular participation to guide educators, policymakers, and mental health professionals in developing targeted programs that support adolescents from diverse socioeconomic backgrounds and promote equitable access to educational resources. The primary achievements of this study include the following points.

1) This study presents a ML-based framework to predict SES using parent-child attachment factors, an underexplored area, by applying seven classifiers and addressing class imbalance through upsampling techniques.
2) Based on model classification outputs, trust, communication and alienation as attachment related dimensions appear to strongly correlate with SES categories and can help inform more precise intervention strategies.
3) This work provides empirical evidence to inform educators and policymakers about the impact of familial relationships on SES, while laying the groundwork for future research incorporating broader socioeconomic indicators and more advanced ML approaches.

II. Literature Review

The relationship between SES, parent–child attachment, and adolescent development has been well documented. SES, encompassing parental education, income, and occupation, directly affects adolescents' access to cognitive resources and academic opportunities.

Li et al. [4] showed that low SES undermines learning conformity by lowering self-esteem, while Rakesh et al. [2] linked financial stress to limited enrichment activities and academic disparities. Conger and Donnellan [3] emphasized that SES indirectly affects academic achievement by introducing environmental instability. Parent–child relationships serve as key mediators. Secure attachment has been associated with higher academic motivation and resilience [9] [10], whereas economic hardship often correlates with harsh parenting and relational instability. Fearon et al. [12] found that insecure attachments in low-SES families predicted externalizing behaviors, while Bachman et al. [22] reported that maternal instability negatively affected socioemotional outcomes. Beelmann and Schmidt-Denter [13] further noted that financial strain weakens post-divorce parent–child interactions, reinforcing disadvantage across generations. Parenting style is another significant factor. Authoritative parenting, characterized by warmth and consistency, supports positive developmental outcomes. In contrast, economic pressure can reduce parental responsiveness. Gibson-Davis and Gassman-Pines [17] emphasized ethnic variations in these effects, and Leidy et al. [20] found that positive parenting among Latino families buffered adolescents from low-SES risks. Extracurricular engagement and peer relationships offer additional resilience. Gao et al. [18] highlighted the role of creative activities in enhancing emotional regulation and problem-solving skills. Chen et al. [19] found that peer attachment mitigates the impact of family instability, while mentorship and community programs strengthen youth development in low-income settings.

Although prior research has examined individual factors like attachment or parenting, few studies have integrated these with extracurricular variables to predict SES. This study addresses that gap by leveraging ML to assess the impact of familial and social dynamics on SES classification.

III. Methodology

This study employs multiple ML algorithms such as Decision Tree, Random Forest, Logistic Regression, SVM, k-NN, AdaBoost, and XGBoost—to examine the link between SES and parent-child attachment. Models were evaluated using stratified 5-fold cross-validation on a preprocessed dataset, which included missing value imputation, outlier removal, normalization, and PCA-based dimensionality reduction. SES class imbalance was addressed via upsampling (Fig. 1).

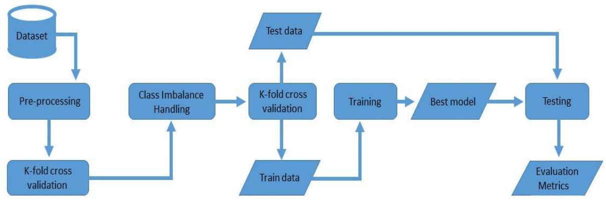

Fig. 1: Proposed Framework for Quantifying SES and Parent-Child Attachment Relationships

A. Dataset

This study utilizes the dataset titled "The Effect of Parent-Child Attachment on Adolescents' Belief in a Just World" [23], comprising 478 samples and 70 attributes from 239 families. Data were collected via structured, private home-based interviews with adolescents and their primary caregivers at two time points (Time 1 and Time 2). The dataset includes key variables on SES, parent-child attachment (trust, communication, alienation for both parents), and adolescents' belief in a just world (BJW). Demographic attributes such as age, sex, school type, and only-child status are also included. SES is categorized into four levels: relatively bad, normal, relatively good, and very good. All psychological measures were obtained using validated psychometric instruments with appropriate adjustments (e.g., reverse scoring). Approval for the study was granted by the Institutional Review Board (IRB), and all participants provided informed consent. To protect confidentiality, data were anonymized and maintained under strict security protocols in line with global ethical requirements.

B. Data Preprocessing

Data preprocessing constitutes a foundational phase in the ML pipeline, as data consistency and quality have a substantial impact on model robustness, accuracy, and generalizability. This subsection outlines all preprocessing steps used in the proposed framework:

- **(a)** *Missing Value Handling:* For categorical variables, missing entries were filled using the mode to retain the most frequent category. Numerical features with missing data were imputed using the median, chosen for its resilience to outliers. These imputation methods preserved the dataset's overall structure while minimizing bias.
- **(b)** *Outlier Detection and Rejection:* The Z-score method was applied to identify outliers, considering data points with $|Z| > 3$ as anomalous. A total of 24 such instances were detected, of which 22 were removed to minimize distortion in distance-based algorithms like KNN and SVM.
- **(c)** *Data Normalization:* All numerical features were normalized using the Z-score method, which rescales the data so that each feature centers around zero with unit variance. This standardization helps models that are affected by differences in feature scales perform more effectively.
- **(d)** *Feature Selection Using PCA:* Dimensionality reduction was performed using Principal Component Analysis (PCA), retaining components capturing 95% cumulative variance. This reduced noise and computational complexity, enabling more efficient and effective model training without significant information loss.

C. Cross-Validation

A stratified 5-fold cross-validation was implemented to assess model robustness and minimize overfitting risk. The dataset was partitioned into five mutually exclusive folds. Each fold iteratively served as the validation set, with the remaining four folds constituting the training partition. The overall performance metric M was derived from the mean of results across all folds, as defined in (1):

$$M = \frac{1}{K}\sum_{n=1}^{K} P_n \pm \sqrt{\frac{\sum_{n=1}^{K}(P_n - \bar{P})^2}{K-1}} \tag{1}$$

where $K = 5$ is the number of folds, P_n is the performance metric on fold n, and $\bar{P}$ is the mean performance across folds.

D. Class Imbalance Handling

Significant class imbalance in the SES target variable distribution, as shown in Fig. 2, was addressed through minority-class upsampling. Underrepresented class instances were replicated until they matched the majority class size, resulting in a balanced distribution that enhanced model performance and consistency across all SES categories.

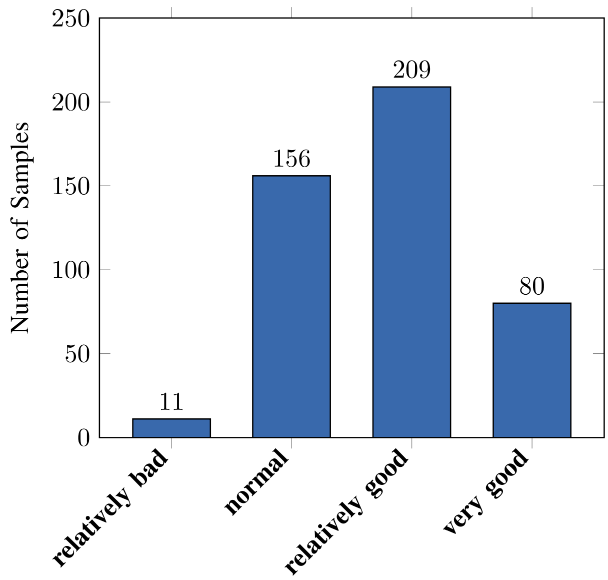

Fig. 2: Class distribution of the SES variable before applying upsampling.

E. Evaluation Metrics

Model performance was assessed using five standard classification metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC. These complementary measures collectively capture distinct performance dimensions to ensure rigorous evaluation.

Accuracy: Accuracy measures the proportion of correctly predicted instances out of all samples tested. It is formally expressed as in (2).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

Precision: Precision is the fraction of positive predictions that are actually correct, reflecting the accuracy of the model's positive classifications. It is calculated mathematically as shown in (3).

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

Recall: Recall, also called sensitivity, represents the percentage of true positive cases that the model successfully detects. It is computed as follows in (4).

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

F1-score: The F1-score is calculated as the harmonic mean of Precision and Recall, reflecting a trade-off between the two metrics to assess overall model effectiveness. It is computed as shown in (5).

$$\text{F1-score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \tag{5}$$

AUC-ROC curve: The AUC-ROC curve is used to show how well a model can separate different classes by comparing the True Positive Rate (TPR) and False Positive Rate (FPR). Both TPR and FPR are defined in (6).

$$\text{TPR} = \frac{TP}{TP+FN}, \quad \text{FPR} = \frac{FP}{FP+TN} \tag{6}$$

In the above equations, TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

F. Implementation Details

All computational experiments were carried out on Kaggle's cloud-based platform, which provided a consistent environment for data preprocessing, model training, validation, and result visualization. The use of Kaggle ensured stable computational resources and supported reproducibility. The platform was accessed via a local machine running Windows 11 Pro.

G. ML Models

To quantify the relationship between SES and parent–child attachment factors, a range of ML models was applied. These included interpretable algorithms as well as advanced approaches such as ensemble and kernel-based methods. Each model's effectiveness was measured according to the previously described standard metrics.

Decision Tree (DT): A supervised learning method that constructs a tree-like model by recursively splitting the data to maximize information gain or minimize Gini impurity. DT was selected because it is easy to interpret and effectively illustrates the impact of attachment factors on SES.

Random Forest (RF): This approach builds multiple decision trees using different random samples of the data and selected features. By aggregating their predictions, RF improves accuracy and helps prevent overfitting, especially when handling complex datasets with numerous variables.

XGBoost: An advanced boosting method that creates decision trees sequentially, with each new tree focusing on correcting errors made by earlier ones. It incorporates both L1 and L2 regularization to reduce overfitting. XGBoost is particularly well-suited for imbalanced datasets and modeling intricate, non-linear interactions within the data.

Support Vector Machine (SVM): A classifier that tries to find the largest gap between different classes. It uses kernel methods to capture complex and non-linear relationships in the data. SVM works especially well when dealing with datasets that have many features and when subtle patterns need to be identified.

Logistic Regression (LR): A statistical method that predicts the probability of an input belonging to a particular class by applying a logistic function to a weighted sum of features. Due to its straightforwardness and ease of interpretation, it is frequently used as a baseline model in data analysis.

k-Nearest Neighbors (k-NN): A straightforward technique that classifies a data point by considering the most common class among its closest neighbors in the feature space. This method was used to identify non-linear patterns and hidden clusters within the attachment data.

AdaBoost: A boosting technique that iteratively emphasizes misclassified instances to improve weak learners. AdaBoost enhances sensitivity to complex SES variations linked to attachment patterns.

IV. Results and Disscusion

This section presents the findings from using ML techniques to explore the relationship between SES and adolescent parent–child attachment. The findings are organized into four parts for clarity. Subsection A outlines the performance of various classifiers before and after addressing class imbalance. Subsection B presents a visual comparison of model accuracies, emphasizing the enhanced effectiveness of ensemble techniques. Subsection C presents the confusion matrix of the optimized XGBoost algorithm to evaluate prediction accuracy for each class. Subsection D discusses how these findings align with existing literature, their practical implications for social science and policy, and methodological limitations to be addressed in future work.

A. Performance Results

This study applied multiple ML models to assess the connection between parent–child attachment and the SES of adolescents. The original imbalanced dataset was used for initial evaluation through 5-fold cross-validation. As shown in Table I, RF achieved the highest baseline accuracy (52.41%), while AdaBoost yielded the lowest (37.95%), highlighting the challenge of SES classification under class imbalance. To mitigate this issue, we employed an upsampling technique to synthetically balance the class distribution. Post-augmentation, all models were retrained using 5-fold cross-validation. Performance improved substantially across all models, as shown in Table II. XGBoost, which initially achieved 50.89% accuracy, reached 89.00% after upsampling, outperforming all other classifiers. DT and RF also exhibited notable gains, achieving 87.33% and 88.40% accuracy, respectively. SVM reached 82.54%, demonstrating effectiveness in complex decision boundaries. AdaBoost improved to 59.20%, reflecting the benefits of class balancing. In addition to accuracy, notable improvements were seen in precision, recall, and F1-score, demonstrating greater robustness and better generalization. These results underscore the importance of handling class imbalance when developing fair and reliable predictive models. Overall, XGBoost demonstrated the highest accuracy in classifying SES, highlighting its effectiveness in managing structured and high-dimensional datasets.

TABLE I: Model performance before handling class imbalance

Algorithm	Average Accuracy (%)
DT	40.34
k-NN	46.71
SVM	47.36
AdaBoost	37.95
XGBoost	50.89
LR	41.24
RF	52.41

TABLE II: Model performance after handling class imbalance (in %)

Algorithm	Accuracy	F1 Score	Precision	Recall	AUC-ROC
DT	87.33	87.20	87.58	87.32	91.55
k-NN	70.34	70.43	71.34	70.26	89.89
SVM	82.54	82.29	82.61	82.50	95.71
AdaBoost	59.20	58.84	59.27	58.89	79.85
XGBoost	**89.00**	**88.94**	**89.29**	**89.02**	**98.40**
LR	69.62	69.21	69.27	69.61	86.50
RF	88.40	88.37	88.60	88.39	98.37

B. *Result Comparison of ML Models*

This subsection compares the classification accuracy of several ML models. As shown in Fig. 3, the XGBoost classifier achieves the highest accuracy at 89.00%, reflecting its strength in modeling complex data patterns. RF performs closely behind with an accuracy of 88.40%, underscoring the effectiveness of ensemble learning techniques. In contrast, AdaBoost records the lowest accuracy of 59.20%, indicating it may be less suited to the dataset's characteristics. LR and KNN both attain moderate accuracy levels near 70%. SVM achieves 82.54% accuracy, demonstrating strong performance though slightly trailing the ensemble methods. Overall, the results indicate that ensemble approaches, particularly XGBoost and RF, deliver improved prediction accuracy by aggregating the predictions of several decision trees. In comparison, simpler models such as LR and KNN show relatively lower accuracy, likely due to the dataset's complexity and non-linear feature interactions.

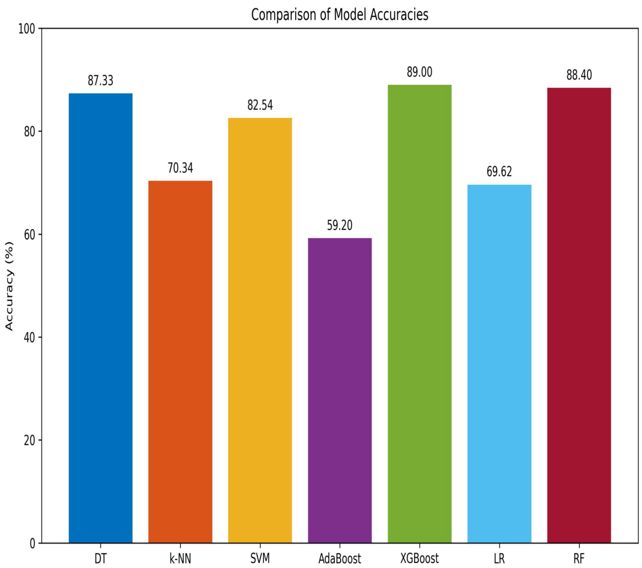

Fig. 3: Accuracy Comparison of Different Models.

C. *Confusion Matrix*

The confusion matrix of the optimized XGBoost algorithm demonstrates strong class-wise predictive performance, as shown in Fig. 4 . High accuracy is observed for distinguishing optimal conditions, especially in high-activity and low-default classes. Misclassifications primarily occur between adjacent categories such as ”very good” and ”relatively good.”

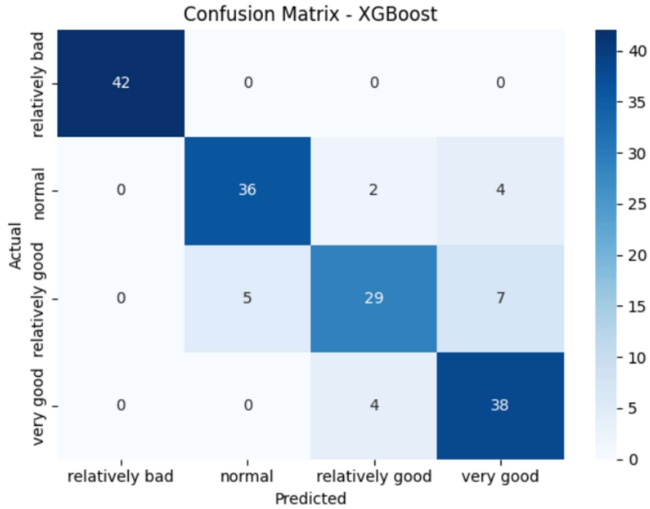

Fig. 4: Confusion matrix for the optimized XGBoost classifier.

D. *Discussion*

This research highlights how ML techniques can be used to measure the link between parent–child attachment and SES in adolescents. Previous studies have shown that insecure attachment styles are frequently linked to lower SES. Our findings align with such literature, reinforcing that attachment dimensions such as trust, communication, and alienation hold predictive value for SES, as supported by Jago [24] and similar studies. The effectiveness of ensemble ML models such as XGBoost and RF highlights their capacity to capture complex, non-linear relationships in psychosocial data, consistent with approaches used by Herd et al. [25]. These models outperform traditional methods by identifying latent structures and interdependencies, making them suitable for behavioral data analysis. The findings carry practical implications. Family-based interventions that improve attachment quality may serve as potential tools for addressing SES-related disparities. Additionally, school-based counseling programs could play a role in strengthening parent–child dynamics.

However, this study has limitations. The sample size (239 families) and limited geographic representation reduce generalizability. The cross-sectional nature of this study restricts conclusions regarding causal relationships. Also, dependence on self-reported information could lead to biased responses. While class imbalance was addressed through upsampling, future work could explore alternative techniques such as SMOTE or ensemble resampling. Hyperparameter tuning was not deeply optimized; methods like grid search or Bayesian optimization may improve performance. Furthermore, causal models (e.g., causal forests) could provide deeper insight into the SES–attachment link.

V. Conclusion

This research developed a detailed ML framework aimed at quantifying the connection between SES and parent–child attachment among adolescents. Rigorous preprocessing including missing value imputation, outlier removal, normalization, and PCA-based feature selection ensured data quality and minimized bias. Experimental results demonstrated the importance of handling class imbalance. Prior to upsampling, models performed suboptimally with RF achieving 52.41% accuracy and XGBoost 50.89% accuracy. Post-oversampling, performance improved significantly across all classifiers, with XGBoost achieving 89.00% accuracy and a 98.40% AUC-ROC, highlighting the effectiveness of ensemble methods in modeling complex socio-behavioral interactions. The results suggest that attachment dimensions such as communication, trust, and alienation may play influential roles in predicting SES, as reflected by their presence among the selected features following PCA. Increased communication and trust corresponded with higher SES categories, such as relatively good and very good, while alienation correlated with lower SES categories. These findings provide empirical evidence for targeted interventions aimed at improving adolescent well-being through enhanced parent–child relationships. From an applied perspective, this framework offers valuable insights for psychologists, social scientists, and policymakers aiming to address socioeconomic disparities using data-driven approaches. Future research may explore deep learning integration with real-time behavioral data to enhance predictive performance and social impact.

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